

Code:

```
#include <stdio.h> #include <stdlib.h>

// Comparison function for ascending order (girls) int cmp_asc(const void *a, const void
b) {return ((int *)a - *(int *)b); }

// Comparison function for descending order (boys) int cmp_desc(const void *a, const
void b) {return ((int *)b - *(int *)a); }

int main() { int t; if (scanf("%d", &t) != 1) return 0;

while (t--) {
    int n;
    if (scanf("%d", &n) != 1) break;

    int a[n], b[n];

    // Read heights of girls
    for(int i=0; i<n; i++) {
        scanf("%d", &a[i]);
    }

    // Read heights of boys
    for(int i=0; i<n; i++) {
        scanf("%d", &b[i]);
    }

    // Sort girls in ascending order
    qsort(a, n, sizeof(int), cmp_asc);

    // Sort boys in descending order
    qsort(b, n, sizeof(int), cmp_desc);

    int ideal_pairs = 0;

    // Check matching condition for each pair
    for(int i=0; i<n; i++) {
        if (a[i] % b[i] == 0 || b[i] % a[i] == 0) {
            ideal_pairs++;
        }
    }
}
```

```
    printf("%d\n", ideal_pairs);  
}  
  
return 0;  
  
}
```

Input:

```
2  
4  
1 6 9 12  
4 12 3 9  
4  
2 2 2 2  
2 2 2 2
```

Output:

```
2  
  
4
```